

LAUNCH CANADA 2026 PAYLOAD CHALLENGE



Final Payload Design Report for Polaris
Competing Team 59 - Waterloo Rocketry

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1 PAYLOAD

The scientific payload of Polaris is a sensor suite focused on vibrational analysis. It includes 8 breakout boards, each equipped with a KX132-1211 three axis accelerometer, and a main board featuring an STM32H7 microcontroller and SHT31 temperature and humidity sensor.

1.1 Experiment Motivation

The payload subsystem has spent much of this year's design cycle designing a fibre optic gyroscope (FOG) for precise measurements of rocket roll rate. FOGs rely on a principle known as the Sagnac effect [1]. When light travels in a loop, such as a path defined by lenses or fibre optic, it takes a certain amount of time to reach a set endpoint. If that loop is rotating, the light experiences centrifugal acceleration and is now in a non-inertial reference frame [2]. This results in a change in the length of the path that the light must travel.

If two identical beams of light counter-propagate around a rotating loop, they can be recombined to produce interference patterns due to the phase differences created by the change in path lengths [1, 3]. Figure 1.1 shows the change in the location of M , where M is the initial starting location of the light and M' is the location after some rotation.

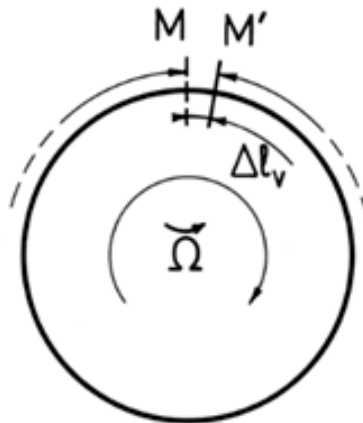


Figure 1.1: Visual representation of the path difference induced by a rotating coil
[1]

The waves undergo a phase shift proportional to the angular velocity of the ring, which changes the power read by a photodiode. This power is a sinusoid with respect to the phase shift, and can be biased by changing the length of the fibre optic using a piezoelectric crystal. An ideal bias of $\pi/2$ provides a maximum sensitivity and the ability to determine direction of rotation.

Due to the prohibitive cost of fibre-coupled optics (i.e. the optical components used to create the Sagnac effect all have built-in fibre optic connectors), the project relies on coupling discrete optics through air. This means the components require very tight tolerances to effectively focus high-power light through the fibre optic coil and into the detector. Slight deviations will lead to losses, particularly when focusing light into the fibre optic. The specialized near-infrared cable has a core diameter of only $80\mu\text{m}$. These tight tolerances demand a well-tested system for damping

the vibrations the optics experience to get high quality data. As will be explored in the further sections, this payload tests the vibrations of several different mounting configurations to determine what damping is ideal for the mounting of the FOG on a future rocket.

1.2 Electrical System

The electrical system consists of one Vibration Logging Board (VLB) and 8 identical Accelerometer Breakout boards (ABBs). Engineering requirements were set to guide the design of these boards. See Table 1.1 for Vibration Logging Board requirements and Table 1.2 for Acceleration Breakout Board requirements.

1.2.1 Electrical Requirements

Requirement	Motivation
Must support the connection of 8 accelerometers	Number of feasible breakout boards to fit in allocated space.
Must support sampling of 8 accelerometers at a minimum rate of 10kHz.	The range from 0Hz to 5kHz provides a comprehensive range of expected vibrations that could displace the optics.
Must support SPI communication.	Fast communication protocol for high speed sampling, widely used within the Waterloo Rocketry team.
Must support internal logging.	Data must be recorded in a manner that is processable post-flight.
Must fit within a 60mm x 100mm x 50mm bounding box.	System integration constraints.
Must run off of 5V with a maximum current draw of less than 1 amp, when integrated with the breakout boards.	Power budget and available power rails.

Table 1.1: Vibration Logging Board Requirements

1.2.2 Accelerometer Breakout Board Design

The eight breakout boards are all identical, featuring the KX132-1211 accelerometer and a JST connector. This particular accelerometer was chosen due to its low cost, 3-axis sampling, high speed communication protocols, and high bandwidth.

This experiment is particularly demanding in terms of sample speed, since even small displacements caused by vibrations of frequencies in the kilohertz can affect the amount of light that gets into the core of the fibre optic. With an output data rate of 25.6kHz, this accelerometer can reconstruct vibrations in all three axes up to over 10kHz in frequency. Additionally, each sample is 16 bits, therefore minute accelerations can be recorded at this speed.

Requirement	Motivation
Must be powered by Vibration Logging Board	Harnessing logistics.
Must contain an accelerometer with an output data rate of at least 10kHz.	Data collection and resolution.
Must support SPI communication.	Fast communication protocol for high speed sampling, widely used within the Waterloo Rocketry team.
Must keep all surface components to one face on the board.	Allows boards to be mounted flat and bonded to flat surfaces.
Must run off of 5V with a collective maximum current draw of less than 1 amp, when integrated with Vibration Logging board.	Power budget and available power rails.

Table 1.2: Accelerometer Breakout Board Requirements

To communicate with the main vibration board, two 10MHz 4-wire SPI buses each with four chip select wires are used. This will provide sufficiently short clock cycles to keep up with the high output data rate.

1.2.3 Vibration Logging Board Design

The Vibration Logging Board contains the microcontroller and the humidity and temperature sensor. This board is responsible for logging all information onto a micro SD card and communicating with the other electronics via the RocketCAN bus. Three strategies are used to maximize the number and consistency of samples recorded. The first is using an STMH7, which features a 480 MHz clock rate for rapid sampling. Additionally, to account for SD card write latency, a ping-pong buffer was implemented in the firmware. This is a dual buffer system in which one buffer is written to by the direct memory access controller (DMA), in parallel with the other buffer being read to the SD card by FatFs. See Figure 1.2 for a visualization. Finally, the last optimization is using 4-bit SDIO line instead of SPI to write to the micro SD card.

Using SDIO for SD card communication over SPI is favourable for this experiment for several reasons. Firstly, SDIO has its own transmission lines from the MCU to the SD card. This is useful since it allows the MCU to communicate with the accelerometers and the SD card faster (since it does not have to switch from SPI slave to SPI slave). The most important reason SDIO is favoured instead of SPI is that SDIO has 4 data transmission lines, allowing for read/write speeds up-to 200Mbps, whereas SPI can transmit up to 20Mbps ([4]). Higher speeds are favoured because we will be recording new data from accelerometers thousands of times per second. These faster transmission speeds allow the MCU to spend less time writing to the SD card, lowering the possibility that data is missed from the accelerometers.

As outlined in Figure 1.2 the system is composed of 3 main parts, SPI, DMA and SDIO. The system begins logging when it recognizes the 'Injector Valve Open' CAN command. Each sample frame contains a timestamp, a temperature reading, and high and low data bytes for the eight sets of X, Y, and Z acceleration readings.

The temperature sensor will be used for analyzing potential impacts on the FOG electronics due to change in ambient environment. The piezoelectric crystal used for the phase bias of the

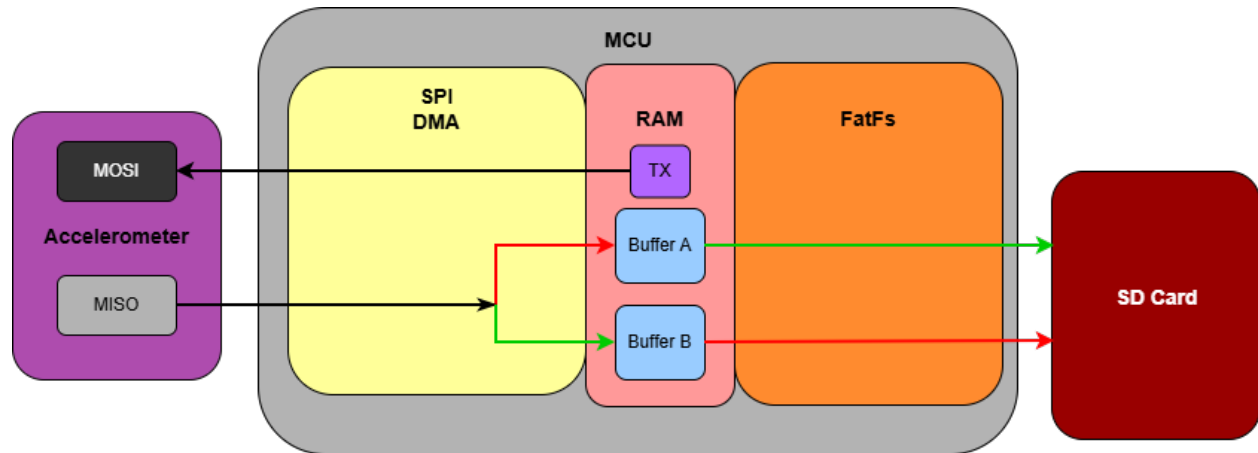


Figure 1.2: Ping pong buffer architecture. The red and green pathways are never active at the same time.

FOG needs to expand to a specific diameter based on the voltage applied. However, when the temperature increases, the diameter tends to oscillate more. Additionally, the laser power electronics must dissipate a fair amount of heat due to the significant current draw of the laser. If ambient temperature rises too much, the dissipation will fail and the laser may no longer draw consistent power. As the temporal resolution of the temperature signal is much lower than that of rocket vibrations, this sensor does not have as strict speed constraints as the accelerometers, and can be sampled less frequently.

1.3 Mechanical System

The mechanical system of the experiment consists of the Payload mounting plate, the weighting plates, and the standoffs that are being tested. Requirements for the mechanical system are outlined in Table 1.3, and descriptions of the various selected standoffs can be found in Table 1.4.

As the breakout boards were much lighter than the expected mass for planned optical experiments, each board was bonded to a small aluminum plate (weighting plates) to increase mass. This is to decrease the impact of noise from very high frequency/low amplitude vibrations that will not affect a heavier experiment with a lower natural frequency and more inertia. Bonding was added in addition to fasteners to increase stiffness of the board-plate interface and minimize vibration between these components that would shroud vibrations from the rocket.

Some types of standoffs or mounting configurations are duplicated. For instance, two sets of control accelerometers will be flown. The first is mounted closer to the center of the rocket, to measure the central vibrations. The other is mounted closer to the edge of the rocket to measure the peripheral vibrations. This results in a control vibration profile that is of a more consistent accuracy for all breakout boards, regardless of their mounting locations relative to fasteners or rocket axes. Additionally, two versions of O-Rings will be flown to compare the impacts of O-Ring stiffness on experiment results, as well as two lengths of metal standoffs.

1.3.1 Mechanical Requirements

Requirement	Motivation
Must be mounted in Canards section [5]	Integration with flight system
Must support the mounting of 8 ABBs and one VLB	Number of feasible breakout boards to fit in allocated space.
Should minimize the use of fasteners (with the exception of the standoffs being tested)	Minimize effect of component chatter on vibration measurements.
Should have sufficient mass to be representative of potential optical experiments	Mass has a significant impact on natural frequency and impact of vibrations on a mechanical system.
Must weigh less than 500g	Rocket mass budget constraints.

Table 1.3: Vibration Logging Board Requirements

1.3.2 Experiment Configuration and Standoff Selection

Position	Breakout Board Label	Standoff Description	Motivation
A	Central Control	No standoffs, mounted and bonded flat to mounting plate	Experiment control, accounts for central vibration profile.
B	Hard O-Ring	Buna-N O-Ring, Durometer 90A	Stiffer option of easily accessible vibration damping standoff.
C	5mm Metal	5mm male-female threaded hex standoff	Most common PCB standoff used by the Waterloo Rocketry team.
D	3D printed stand-off	5mm PETG clearance hole standoff	Customizable alternative to metal standoffs.
E	RTV	5mm male-female threaded hex standoff	Most common PCB standoff used by the Waterloo Rocketry team.
F	Soft O-Ring	Buna-N O-ring, Durometer 50A	Less stiff options of easily accessible vibration damping standoff.
G	8mm Metal	8mm male-female threaded hex standoff	Longer version of common PCB standoffs used by the Waterloo Rocketry team.
H	Peripheral Control	No standoffs, mounted and bonded flat to mounting plate	Experiment control, accounts for peripheral vibration profile.

Table 1.4: Standoff Configuration

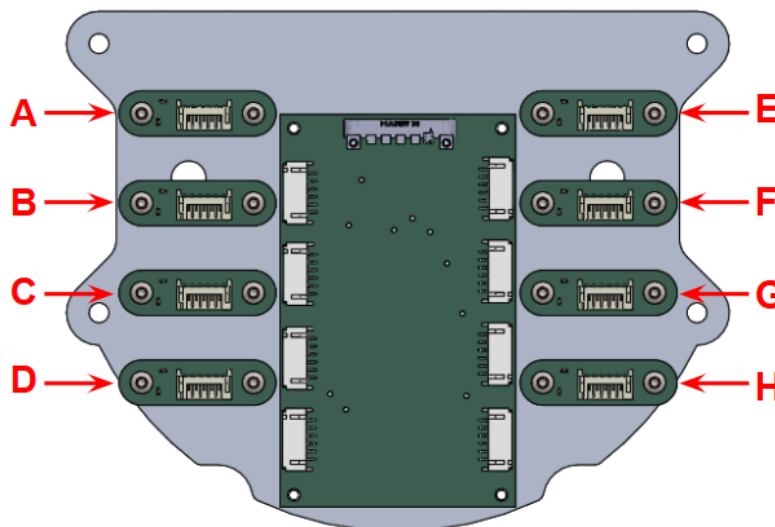


Figure 1.3: Breakout board configuration

1.4 Testing

All components that will make up the payload will be tested individually, then tested in combination with the systems until at the flight ready state. The main testing includes the following:

- Individual PCB tests. This will evaluate that all hardware and firmware on the boards are working as intended, and verify electrical requirements listed in Section 1.2.1.
- Assembly testing. All machined components will be fit checked to ensure all mechanical requirements, listed in Section 1.3.1 have been validated.
- Flight-like assembly and testing. This will test that all components fit together and that the accelerometers work as intended. The communication both to the internal SD card and CANBUS will be assessed.

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